Noise-Robust Tolls in Atomic Congestion Games

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Abstract: We consider a congestion game with a finite number of agents selecting paths according to a logit dynamics with fixed noise level. We propose an iterative distributed tolling scheme and prove its convergence to a toll vector that minimizes the expected total travel time for every noise level.

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1. INTRODUCTION

We consider a congestion game with a finite number of agents selecting paths according to a logit dynamics with fixed noise level. When the population is infinitely large, marginal cost tolls can restore the social optimum under the assumption of perfect rationality (Beckmann et al. (1956)), but these assumptions rarely hold. In fact, in presence of noise, the invariant distribution of the logit dynamics concentrates, as the number of agents grows large, on perturbed equilibria, which are in general suboptimal (Sandholm (2011)). In this work, we establish an iterative distributed algorithm for toll design in congestion games to minimize the expected total travel time on the network for every noise level, when the number of players is large and finite. This work is related to (Park and Leonard (2025)), where an incentive mechanism that guarantees convergence to desirable configurations was provided for every noise level in contractive games with a continuum of users.

Notation: Let Δ and $\Delta_n = \Delta \cap \frac{1}{n}\mathbb{Z}_+^m$ denote the m-dimensional simplex and its discrete counterpart, whose size may be deduced from the context. Let $\delta^\theta(x)$ denote the Dirac delta centered in θ.

2. PROBLEM SETTING

We model the network by a directed multigraph $G = (V, E)$. We consider a single origin-destination pair o, d in $\mathcal{V}$, with d reachable from o, and let Γ denote the non-empty set of $o-d$ paths. We define the link-path incidence matrix A in $\{0,1\}^{E\times\Gamma}$ with entries $A_{e\gamma} = 1$ if and only if e in γ. We consider n indistinguishable players, each choosing a path in Γ, and collect the empirical path frequencies into the state vector x in Δ_n, with x_γ denoting the fraction of players choosing γ. The cost of path γ is

$$c_\gamma^\theta(x) = \sum_{e\in E} A_{e\gamma}\left(\tau_e\left((Ax)_e\right) + \omega_e^\theta\left((Ax)_e\right)\right) , \qquad (1)$$

where, for every link e in E, $\tau_e : \mathbb{R}_+ \to \mathbb{R}_+$ is a continuously, differentiable, non-decreasing, positive, and convex delay function that accounts for congestion, and $\omega_e^\theta : \mathbb{R}_+ \to \mathbb{R}_+$ is a monetary *toll* parametrized by θ. Since the cost functions depend on the path distribution, this setting defines a population game.

The population state evolves by the following discrete-time Markov chain $X(t)$ on Δ_n, called logit dynamics. At each time step, an agent is selected uniformly at random and selects a random path drawn with a logit probability distribution. That is, for every state x in Δ_n and paths i, j in Γ with $x_i \neq 0$, the transition probability is

$$P_{xy}^{(n)} = \begin{cases} x_i \dfrac{e^{-\eta^{-1}c_j^\theta(y)}}{\sum\limits_{k\in\Gamma} e^{-\eta^{-1}c_k^\theta(x+\frac{1}{n}(u^k-u^i))}} & \text{if } y = x + \dfrac{1}{n}(u^j - u^i) \\ 0 & \text{otherwise}, \end{cases}$$

where $\eta > 0$ is the noise level and u^i is the vector with 1 in the i-th entry and 0 elsewhere. The Markov chain $X(t)$ is irreducible for every noise level. Hence, it admits a unique invariant distribution π_n, which depends on the toll functions.

The goal of the planner is to design a vector of toll functions that minimize the expected total travel time, i.e.,

$$\min_{\{\omega_e:\mathbb{R}_+\to\mathbb{R}_+\}_{e\in E}} \mathbb{E}_{\pi_n}\left[\sum_{e\in E}\sum_{\gamma\in\Gamma} X_\gamma A_{e\gamma}\tau_e\left((AX)_e\right)\right] . \qquad (2)$$

3. OPTIMAL TOLLS

Let the Total Travel Time $\phi : \Delta \to \mathbb{R}_+$ be defined by

$$\phi(x) = \sum_{\gamma\in\Gamma} x_\gamma \sum_{e\in E} A_{e\gamma}\tau_e\left((Ax)_e\right) , \qquad (3)$$

and

$$\mathcal{X}^* = \arg\min_{x\in\Delta}\{\phi(x)\} .$$

In this work, we shall assume that the network is composed of E parallel links from o to d, so that $\Gamma = E$. Moreover, we assume that $\phi(x)$ is strictly convex, which implies that $\mathcal{X}^* = \{x^*\}$ is a singleton. Furthermore, we shall assume for ease of exposition that x^* lies in $\mathrm{int}(\Delta)$, representing the interior of Δ. We remark that x^* is the flow that minimizes the total travel time when the number of agents is infinitely large.

In the limit of infinitely large population, marginal cost tolls

$$\tilde{\omega}_e(x_e) = x_e \tau_e'(x_e)\,,$$

are known to align the unique equilibrium of the game with the social optimum. However, in presence of noise, these tolls are no longer optimal (Sandholm (2011)). The main idea of this work is to combine marginal cost tolls with noise-dependent tolls

$$\bar{\omega}_e(\theta) = -\eta \log(\theta_e)\ ,$$

with θ in Δ, to address the distortion introduced by the noise. We then propose an iterative distributed algorithm for updating θ in such a way that the toll vector

$$\omega_e^\theta(x_e) = \tilde{\omega}_e(x_e) + \bar{\omega}_e(\theta_e)\,, \quad \forall e \in E\,, \tag{4}$$

converges to the solution of (2) for every noise level $\eta > 0$ and for every initial condition θ^0 in $\mathrm{int}(\Delta)$, in the limit of large population. The parameter θ is updated through a discrete-time dynamical system that evolves on a much slower time scale than the underlying Markov chain.

Let π_n^θ denote the invariant distribution of the Markov chain $X(t)$ with tolls (4). Then, let $\psi : \Delta \to \Delta$ defined by

$$\psi_n(\theta) = \mathbb{E}_{\pi_n^\theta}[X] = \sum_{x \in \Delta_n} x \pi_n^\theta(x)\,,$$

and consider the discrete-time dynamical system

$$\theta^{k+1} = \psi_n(\theta^k)\ .$$

We also define the map

$$\Psi(\theta) = \arg\min_{x \in \Delta} \left\{\eta^{-1}\phi(x) + D_{KL}(x||\theta)\right\}\ ,$$

where $D_{KL}(x|\theta) = \sum\limits_{e\in E} x_e \log\left(\frac{x_e}{\theta_e}\right)$.

It is possible to verify that

$$\psi_n(\theta) \overset{n\to+\infty}{\longrightarrow} \Psi(\theta)\,, \quad \forall \theta \in \Delta_n\,, \tag{5}$$

which prompts the study of the limit dynamical system

$$\theta^{k+1} = \Psi(\theta^k)\ . \tag{6}$$

Proposition 1. The limit dynamical system (6) admits a fixed point $\theta^* = x^*$. Moreover, θ^* is the unique fixed point in $\mathrm{int}(\Delta)$, and

$$\lim_{k\to+\infty} \theta^k = \theta^*\,, \quad \forall \theta^0 \in \mathrm{int}(\Delta)\,.$$

Proof. The proof is deferred to Appendix B.

We can now prove our main result.

Theorem 1. Consider the toll $\omega^\theta(x)$ defined in (4). For every $\epsilon > 0$, there exists k_ϵ in $\mathbb{N}$ and, for every $k > k_\epsilon$, an integer $n_{\epsilon,k}$ in $\mathbb{N}$ such that

$$\mathbb{E}_{\pi_n^{\theta^k}}\Big[\sum_{e\in E} X_e\tau_e(X_e)\Big] - \inf_\pi \mathbb{E}_\pi\Big[\sum_{e\in E} X_e\tau_e(X_e)\Big] < \epsilon.$$

for every $n > n_{\epsilon,k}$.

Proof. The proof is deferred to Appendix D.

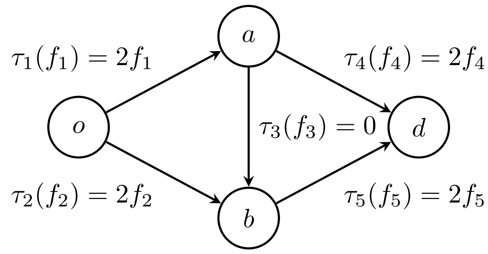

Fig. 1. Wheatstone network.

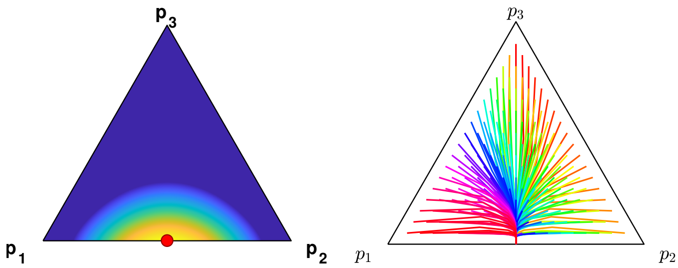

Fig. 2. Left: Invariant distribution of the Markov chain with $n = 500$ players and noise level $\eta = 1$. In red, the social optimum. Right: Trajectories of θ under (6) from various initializations, all converging to $\theta^* = (1/2, 1/2, 0)$.

4. EXAMPLE

In this section we provide an example to show that our results may be extended to cases where the social optimum does not lie in the interior of the simplex and to networks with non-parallel links. Consider the Wheatstone network in Figure 1, with set of paths is $\Gamma = \{(o,a,d),(o,b,d),(o,a,b,d)\}$, which we shall denote by p_1, p_2, and p_3, respectively.

We consider a congestion game with $n = 500$ players and a noise level of $\eta = 1$, and compute the invariant distribution both with and without the noise-dependent tolls. The invariant distribution concentrates around the social optimum, as shown in the left plot of Figure 2. Moreover, in the right hand side of Figure 2 reports the evolution of θ_t initialized at different values of θ, where each sequence converges to $\theta^* = (0.5, 0.5, 0)$.

5. CONCLUSION

We proposed an iterative distributed tolling scheme that steers the invariant distribution of a Markov chain toward the social optimum. Future work will extend it to general networks and boundary optima.

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Appendix A. LEMMA 1

Lemma 1. For every $\epsilon > 0$, there exists n_ϵ such that $|\pi_n^\theta(x) - \delta^{\Psi(\theta)}(x)| < \epsilon$ for every $n > n_\epsilon$ and $x \in \Delta_n$.

Proof. In order to define the invariant distribution, let us first introduce the potential function in the finite population case, i.e.,

$$\phi_n(x) = \sum_{e\in\mathcal{E}} \sum_{j=0}^{n(Ax)_e} \left(\tau_e\left(\frac{j}{n}\right) + \tilde{\omega}_e\left(\frac{j}{n}\right)\right) - n\eta\left(x'\log(\theta)\right),$$

and we defined $\bar{\phi}_n(x)$ as

$$\bar{\phi}_n(x) = \sum_{e\in\mathcal{E}} \sum_{j=0}^{n(Ax)_e} \left(\tau_e\left(\frac{j}{n}\right) + \tilde{\omega}_e\left(\frac{j}{n}\right)\right) .$$

Then, consider two states x and y in Δ_n such that $x = y + \frac{1}{n}(u_i - u_j)$ and compute the probabilities of going from x to y and their ratio, i.e.,

$$\begin{aligned}\frac{P_{yx}}{P_{xy}} &= \frac{y_i e^{\eta^{-1}\left(c_i^\theta(x)\right)}}{x_j e^{\eta^{-1}\left(c_j^\theta(y)\right)}} \frac{\sum_{a\in\Gamma} e^{\eta^{-1}\left(c_a^\theta\left(y+\frac{1}{n}(u_a-u_i)\right)\right)}}{\sum_{a\in\Gamma} e^{\eta^{-1}\left(c_a^\theta\left(x+\frac{1}{n}(u_a-u_j)\right)\right)}} \\ &= \frac{y_i}{x_j}\frac{e^{\eta^{-1}\phi_n(x)}}{e^{\eta^{-1}\phi_n(y)}} ,\end{aligned}$$

where the last equality follows from the definition of potential and the fact that

$$\begin{aligned}x + \frac{1}{n}(u_a - u_j) &= y + \frac{1}{n}(u_i - u_j) + \frac{1}{n}(u_a - u_j) \\ &= y + \frac{1}{n}(u_a - u_i) ,\end{aligned}$$

From the reversibility of the Markov chain it holds that $\frac{\pi_n^\theta(x)}{\pi_n^\theta(y)} = \frac{P_{yx}}{P_{xy}}$, so that it is possible to derive the invariant distribution as

$$\pi_n^\theta(x) = \frac{\binom{n}{nx}\prod_{a\in\Gamma}\theta_a^{nx_a} e^{-\eta^{-1}\bar{\phi}_n(x)}}{\sum_{y\in\Delta_n}\binom{n}{ny}\prod_{a\in\Gamma}\theta_a^{ny_a} e^{-\eta^{-1}\bar{\phi}_n(y)}} .$$

We compute the limit of the invariant distribution as

$$\begin{aligned}\pi^\theta(x) &= \lim_{n\to\infty}\pi_n^\theta(x) \\ &= \lim_{n\to\infty}\frac{\exp(A_n(x))}{Z_n} ,\end{aligned} \quad \text{(A.1)}$$

where

$$\begin{aligned}A_n(x) = &\sum_{k=1}^{n}\log k - \sum_{i=1}^{m}\sum_{k=1}^{nx_i}\log k + n\sum_{a\in E} x_a\log(\theta_a) \\ &-\eta^{-1}\phi_n(x)\end{aligned}$$

and

$$Z_n = \sum_{x\in\Delta_n}\exp(A_n(x)\ .)$$

Notice that the potential $\bar{\phi}_n(x)$ converges to $\phi(x)$ in the limit, i.e.,

$$\begin{aligned}\lim_{n\to+\infty}\frac{1}{n}\bar{\phi}_n\left(\frac{k_n}{n}\right) &= \lim_{n\to+\infty}\frac{1}{n}\sum_{e\in E}\left(\sum_{j=0}^{(Ak_n)_e}\tau_e\left(\frac{j}{n}\right) + \tilde{\omega}_e\left(\frac{j}{n}\right)\right) \\ &= \sum_{e\in\mathcal{E}}\int_0^{(Ax)_E}\left(\tau_e(s) + \tilde{\omega}_e(s)\right)ds \\ &= \sum_{e\in E}\int_0^{(Ax)_e}\left(\frac{d}{ds}\left(s\tau_e(s)\right)\right)ds \\ &= \sum_{\gamma\in\Gamma} x_\gamma \sum_{e\in E} A_{e\gamma}\tau_e\left((Ax)_e\right) \\ &= \phi(x) .\end{aligned}$$

So, we can compute the limit of $A_n(x)$ as

$$\lim_{n\to+\infty} A_n(x) = -n\left(\eta^{-1}\phi(x) - D_{KL}(x||\theta)\right) ,$$

so that

$$\begin{aligned}\pi^\theta(x) &= \lim_{n\to\infty}\frac{\exp(A_n(x))}{Z_n} \\ &= \begin{cases}1 & \text{if } x \in \arg\min_{x\in\Delta}\left\{\eta^{-1}\phi(x) + D_{KL}(x\|\theta)\right\} ,\\ 0 & \text{otherwise}\end{cases} \\ &= \delta^{\Psi(\theta)}(x) .\end{aligned}$$

By applying the definition of limit we conclude that for every $\epsilon > 0$, there exists n_ϵ such that $|\pi_n^\theta(x) - \delta^{\Psi(\theta)}(x)| < \epsilon$, for every $n > n_\epsilon$ and x in Δ_n.

Appendix B. PROOF OF PROPOSITION 1

A fixed point of $\Psi(\theta)$ must inevitably satisfy the following

$$\theta^* = \arg\min_{x\in\Delta}\left\{\eta^{-1}\phi(x) + D_{KL}(x||\theta^*)\right\} .$$

Since $D_{KL}(x||\theta^*)$ diverges whenever x approaches the boundary of Δ, we get that

$$\eta^{-1}\nabla\phi(\theta^*) - \nabla D_{KL}(\theta^*||\theta^*) = c\mathbb{1} ,$$

where $c\mathbb{1}$ follows from the constraint that x is in Δ. Since $\nabla D_{KL}(\theta^*||\theta^*) = 0$, we are left with

$$\eta^{-1}\nabla\phi(\theta^*) = c\mathbb{1} ,$$

which implies that the fixed point of the map is the argmin of ϕ, since ϕ is strictly convex. The existence and uniqueness of an internal fixed point follows from the fact that x^* is stationary for $\phi(x)$, from the assumption that x^* is internal, and from strict convexity of ϕ.

We now prove convergence. Notice that, since $\theta^{k+1} = \Psi(\theta^k)$ and

$$\Psi(\theta) = \arg\min_{x\in\Delta}\left\{\eta^{-1}\phi(x) + D_{KL}(x||\theta)\right\} ,$$

we have

$$\eta^{-1}\phi(\Psi(\theta)) + D_{KL}\left(\Psi(\theta)||\theta\right) \le \eta^{-1}\phi(\theta) + D_{KL}(\theta||\theta) .$$

Since $D_{KL}(\theta||\theta) = 0$, this implies that

$$\phi(\Psi(\theta)) \le \phi(\theta) - \eta D_{KL}(\Psi(\theta)||\theta) .$$

Given that $D_{KL}(\Psi(\theta)||\theta) \ge 0$, then it holds

$$\phi\left(\Psi(\theta)\right) \le \phi(\theta)$$

and therefore ϕ is a Lyapunov function. Hence, LaSalle principle for discrete-time dynamical systems (Mei and Bullo (2017)) ensures convergence to the largest invariant set in Δ. We now aim to exclude convergence to fixed points on the boundary of Δ. The fact that the unique minimizer θ^* of

$$\phi(x) = \sum_{e\in E} x_e \tau_e(x_e) \ ,$$

is in $\mathrm{int}(\Delta)$ implies the existence of $T^* > 0$ such that, for every e in E,

$$\tau_e(\theta^*_e) + \tilde{\omega}_e(\theta^*_e) := T^* \,.$$

Consider now a fixed point $\tilde{\theta}$ in $\partial\Delta$ such that $\Psi(\tilde{\theta}) = \tilde{\theta}$ and define

$$\mathcal{A} = \{e \in E : \theta_e = 0\} \ , \mathcal{B} = E \setminus \mathcal{A} \,.$$

To be a fixed point of Ψ on the boundary, $\tilde{\theta}$ must satisfy

$$\tilde{\theta} \in \underset{x\in\Delta : x_i = 0 \ ,\forall i\in\mathcal{A}}{\arg\min} \sum_{e\in E} x_e \tau_e(x_e) \ .$$

From this, it follows the existence of T, strictly larger than T^*, which corresponds to the cost experienced by agents choosing paths in $\mathcal{B}$, i.e,

$$\tau_e(\tilde{\theta}_e) + \tilde{\omega}_e(\tilde{\theta}_e) := T > T^* \,, \quad \forall e \in \mathcal{B} \,,$$

It also implies that the experienced cost by agents choosing agents in $\mathcal{A}$ is strictly smaller than T^*, i.e.,

$$\tau_e(\tilde{\theta}) + \tilde{\omega}_e(\tilde{\theta}) < \tau_e(\theta^*_e) + \tilde{\omega}_e(\theta^*_e) = T^* \,, \quad \forall e \in \mathcal{A} \,.$$

This implies that, for every i in $\mathcal{B}$ and j in $\mathcal{A}$,

$$\frac{\partial\phi(\tilde{\theta})}{\partial\theta_i} - \frac{\partial\phi(\tilde{\theta})}{\partial\theta_j} = \tau_i(\tilde{\theta}_i) + \tilde{\theta}_i \tau_i'(\tilde{\theta}_i) - \tau_j(\tilde{\theta}_j) - \tilde{\theta}_j \tau_j'(\tilde{\theta}_j) > 0 \,.$$

This implies the existence of a neighborhood O of $\tilde{\theta}$ such that, for every θ in $O \cap \mathrm{int}(\tilde{\theta})$, it holds $\phi(\theta) < \phi(\tilde{\theta})$. Moreover, since $D_{KL}(x\|\theta) = +\infty$ whenever $x_e > 0 = \theta_e$ for some $e \in E$, if $\theta \in \mathrm{int}(\Delta)$ then $\Psi(\theta) \in \mathrm{int}(\Delta)$ as well; hence $\mathrm{int}(\Delta)$ is positively invariant under Ψ, and the trajectory θ^k can never reach $\partial\Delta$ in finitely many steps. Suppose, then, that $\theta^k \to \tilde{\theta} \in \partial\Delta$. For k large enough, $\theta^k \in O \cap \mathrm{int}(\Delta)$, and therefore $\phi(\theta^k) < \phi(\tilde{\theta})$. But ϕ is non-increasing along the trajectory, so $\phi(\theta^{k'}) \leq \phi(\theta^k) < \phi(\tilde{\theta})$ for all $k' > k$, which contradicts $\phi(\theta^{k'}) \to \phi(\tilde{\theta})$. Hence $\tilde{\theta}$ cannot be a limit point of (θ^k).

Appendix C. LEMMA 2

Lemma 2. Given $\epsilon > 0$, let θ^0 in $\mathrm{int}(\Delta)$ and $k > 0$ such that $|\Psi^k(\theta^0) - \theta^*| < \gamma$. Then, there exists $n_{\epsilon,k}$ in $\mathbb{N}$, such that, for every $n > n_{\epsilon,k}$,

$$|\psi^k_n(\theta^0) - \theta^*| < 2\epsilon \,.$$

Proof. From (5), it follows that the discrete map converges to the limit map

$$\psi_n(\theta^0) \overset{n\to+\infty}{\longrightarrow} \Psi(\theta^0),$$

it also holds for a fixed k that

$$\psi^k_n(\theta^0) \overset{n\to+\infty}{\longrightarrow} \Psi^k(\theta^0) \,.$$

This implies that for every $\epsilon > 0$, there exists $n_{\epsilon,k}$ such that for every $n > n_{\epsilon,k}$

$$|\psi^k_n(\theta^0) - \Psi^k(\theta^0)| < \epsilon \,.$$

By combining this with the assumption

$$|\Psi^k(\theta^0) - \theta^*| < \epsilon \,,$$

and using the triangle inequality, we conclude

$$|\psi^k_n(\theta^0) - \theta^*| \leq |\psi^k_n(\theta^0) - \Psi^k(\theta^0)| + |\Psi^k(\theta^0) - \theta^*| < 2\epsilon \,,$$

for every $n > n_{\epsilon,k}$.

Appendix D. PROOF OF THEOREM 1

First, note that, for parallel links,

$$\sum_{\gamma\in\Gamma} X_\gamma A_{e\gamma} \tau_e\left((AX)_e\right) = X_e \tau_e(X_e) \,.$$

Since the delay functions τ_e are continuous, it follows that, for every $\epsilon > 0$, there exists $\sigma_\epsilon > 0$ such that, for every $\theta \in \Delta$ with $|\theta - \theta^*| < \sigma_\epsilon$,

$$|\mathbb{E}_{\delta^\theta}[X_e\tau_e(X_e)] - \mathbb{E}_{\delta^{\theta^*}}[X_e\tau_e(X_e)]| < \epsilon/2 \,. \tag{D.1}$$

From Proposition 1, it follows the existence of k_ϵ such that

$$|\Psi^k(\theta^0) - \theta^*| < \sigma_\epsilon/2 \,, \quad \forall k > k_\epsilon \,. \tag{D.2}$$

Hence, from Lemma 2, for every $k > k_\epsilon$, there exists $n_{\epsilon,k}$ such that

$$|\psi^k_n(\theta^0) - \theta^*| < \sigma_\epsilon \,, \quad \forall n > n_{\epsilon,k} \,.$$

By the choice of σ_ϵ above, this implies

$$|\Psi(\psi^k_n(\theta^0)) - \theta^*| < \sigma_\epsilon/2 < \sigma_\epsilon, \quad \forall n > n_{\epsilon,k}. \tag{D.3}$$

Moreover, by Lemma 1 applied at $\theta = \psi^k_n(\theta^0)$, there exists $\tilde{n}_\epsilon$ such that

$$\left|\mathbb{E}_{\pi_n^{\psi^k_n(\theta^0)}}[X_e\tau_e(X_e)] - \mathbb{E}_{\delta^{\Psi(\psi^k_n(\theta^0))}}[X_e\tau_e(X_e)]\right| < \epsilon/2 \tag{D.4}$$

for every $n > \tilde{n}_\epsilon$. Using (D.4), (D.1) with $\theta = \Psi(\psi^k_n(\theta^0))$, from (D.3), and the triangle inequality, we get

$$\left|\mathbb{E}_{\pi_n^{\psi^k_n(\theta^0)}}[X_e\tau_e(X_e)] - \mathbb{E}_{\delta^{\theta^*}}[X_e\tau_e(X_e)]\right| < \epsilon,$$

for every $n > \max(n_{\epsilon,k}, \tilde{n}_\epsilon)$
Recall now that $\theta^* = x^*$, where

$$x^* = \underset{x\in\Delta}{\arg\min}\{x_e\tau_e(x_e)\} \,.$$

Hence,

$$\delta^{\theta^*}_n = \underset{\pi}{\arg\min}\{\mathbb{E}_\pi[X_e\tau_e(X_e)]\} \,.$$

This concludes the proof.